Journal of Autism and Developmental Disorders (2026) 56:1808–1819
https://doi.org/10.1007/s10803-024-06682-y

ORIGINAL ARTICLE

Peer Problems and Prosocial Behavior Among Icelandic Children and Adolescents with ADHD and/or Autism: Gender and Age Differences

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Accepted: 1 December 2024 / Published online: 21 December 2024
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Abstract

Children with neurodevelopmental disorders tend to have more social difficulties than typically developing children. The aim of the current study was to examine parent and teacher-reported effects of age and gender on social functioning in a large clinical sample of children and adolescents with ADHD, autism, or co-occurring ADHD and autism using a cross-sectional study design. This nationwide clinical sample included 2132 Icelandic children and adolescents (35% girls, 65% boys) aged 5–18 years referred for a neurodevelopmental diagnostic assessment (ADHD and/or autism) in Iceland. Social functioning was measured using the Prosocial behavior and Peer problem subscales on the Strengths and Difficulties Questionnaire (SDQ) completed by parents and teachers. Results revealed that autistic youth and youth with co-occurring ADHD and autism experienced more peer problems and showed less prosocial behavior than youth with ADHD only. According to parents and teachers, girls were found to experience more social difficulties compared to boys. Interaction for age and gender, although only significant for teacher reports, indicated that younger girls with neurodevelopmental disorders experience more peer problems and show less prosocial behavior than older girls. In contrast, boys with neurodevelopmental disorders experience similar issues at all ages. The results suggest different patterns of social difficulties for boys and girls with neurodevelopmental disorders. Future research should examine different developmental pathways of social challenges for boys and girls. Implications for developing and providing clinical interventions appropriate developmental stages are discussed.

Keywords Attention-deficit/hyperactivity disorder (ADHD) · Autism · Youth · Strengths and difficulties questionnaire · Social functioning · Gender differences

Autism spectrum disorder (hereafter called autism) and Attention-deficit/hyperactivity disorder (ADHD) are two of the most common neurodevelopmental disorders among children and adolescents (Francés et al., 2022) and frequently co-occur (Lai et al., 2019; Rommelse et al., 2010). Social difficulties are a transdiagnostic feature for children with ADHD and autism and are central to both disorders (Leitner, 2014), although their difficulties may vary in nature (Mikami et al., 2019). Regarding youth with ADHD, it has been suggested that attention problems may limit their opportunities to notice social cues and demonstrate appropriate social skills (Mikami et al., 2019). Impulsive and hyperactive children may be considered intrusive by peers, which may lead to rejection by the group (Hoza, 2007; Ros & Graziano, 2018). Overall, research has shown that youth with ADHD are more likely to have problems socially, are liked less compared with peers without ADHD, and have

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fewer friends (Gardner & Gerdes, 2015; Hoza et al., 2005; Hoza, 2007; Mikami & Lorenzi, 2011; Spender et al., 2023). Gender differences have also emerged in studies on social functioning among children with ADHD. Boys with ADHD tend to show more noncompliance than boys without ADHD but show no difference in the frequency of prosocial behavior, such as taking initiative socially, showing leadership, resolving conflicts, and setting social norms (Erhardt & Hinshaw, 1994). However, ADHD has been suggested to be somewhat more impairing for girls than boys. Hyperactivity and impulsivity symptoms may be considered more inappropriate in girl groups, causing girls to experience peer rejection, long-lasting social problems, and increased rates of co-occurring psychiatric disorders (Mikami et al., 2013; Mikami & Lorenzi, 2011; Ragnarsdottir et al., 2018; Tung et al., 2016).

In contrast to the social difficulties experienced by children with ADHD, limited prosocial behavior and social motivation may partly be driving the social difficulties experienced by autistic children in terms of making friends and maintaining friendships (Mikami et al., 2019). It has been suggested that diminished social interest in early childhood may reduce social input and chances to learn for autistic youth, which may, in turn, contribute to their difficulties in social communication. Autistic children have been found to engage in fewer verbal interactions and less social play than typically developing children (Corbett et al., 2014). It has been hypothesized that a low sense of reward experienced from social interaction diminishes the motivation to initiate and sustain social behavior, potentially resulting in a reluctance to perform prosocial behaviors (Chevallier et al., 2012). Research has also shown that autistic youth are more likely to be isolated across all grade levels and that they become more isolated in later grades (Rotheram-Fuller et al., 2010), indicating that older autistic children may face more social difficulties than younger children. However, not all autistic children lack social motivation. Some seek out social interactions with peers but may struggle to demonstrate appropriate social behaviors (Scheeren et al., 2012).

Some gender differences have been noted in studies on social interactions among autistic youth. Studies have shown that autistic boys are more likely to be excluded socially by peers. Autistic girls are, on the other hand, neither accepted nor rejected but seem to be ignored or overlooked (Dean et al., 2014). It has been suggested that autistic girls may blend in more efficiently, while boys are more prone to isolation (Hiller et al., 2014). According to the camouflage hypothesis (Lai et al., 2011), autistic females may use compensatory behaviors to mask their social challenges and use social opportunities to watch the interactions of other girls to mimic them. This imitation of appropriate social behavior may help them blend in with peers (Attwood, 2008; Cook et al., 2021; Dean et al., 2017). It is important to note that camouflaging seems easier for those with better executive functioning skills, irrespective of gender (Fletcher-Watson & Happé, 2019; Hull et al., 2021).

Research on social behavior and functioning among children diagnosed with co-occurring ADHD and autism has shown mixed results. However, some studies indicate greater impairment in social and emotional cognitive processes when children are diagnosed with both disorders (e.g., Antshel et al., 2016; Rosello et al., 2022). It has been suggested that if social behavior patterns switch between aloofness (e.g., in autism) and intrusiveness (e.g., in ADHD), the unpredictability in response might cause increased problems in social situations (Mikami et al., 2019). Some studies have shown that children with co-occurring ADHD and autism face more social challenges than children diagnosed only with autism (Factor et al., 2017; Rao & Landa, 2014; Sikora et al., 2012). Other studies have, however, shown no differences in peer problems when comparing boys with co-occurring ADHD and autism to boys with either ADHD or autism. Boys with the combined diagnoses have been found to show less prosocial behavior than boys with ADHD only. However, no difference emerged for boys with the combined diagnosis and boys with autism (e.g., Yamawaki et al., 2020). Similarly, a recent study found comparable social difficulties for autistic children compared with children who have co-occurring ADHD and autism (Harkins et al., 2022). Unfortunately, almost no studies have been conducted on the social functioning of girls with co-occurring ADHD and autism (Mikami et al., 2019). Therefore, information is lacking on how girls with ADHD and autism fare socially and whether it is necessary to provide different clinical interventions based on age and gender for this group of youth with co-occurring neurodevelopmental disorders.

Additionally, most previous studies on social functioning have relied on parent reports, but very few studies have provided teacher reports of social behaviors among youth with ADHD and/or autism (Mikami et al., 2019). The field lacks information on how youth with co-occurring ADHD and autism are viewed socially in the school environment. While obtaining information on social behaviors from parents and teachers is informative, discrepancies between these two types of informants frequently occur (De Los Reyes et al., 2023). Discrepancies in multi-informant information can however provide valuable and relevant information on how behavior presents in a certain context, have high internal consistency, correlate with the perspectives informants have on the children they are assessing and relate to variations in the context where the informants are observing the children (De Los Reyes, 2011). Since this clinical group of children with co-occurring ADHD and autism is underresearched, information from both parents and teachers on

this group should add meaningfully to the current literature. Since social functioning is a central part of both ADHD and autism, providing more comprehensive research on this crucial aspect of children´s lives is essential and valuable for the field of clinical child psychology.

The purpose of the current study was to examine parent and teacher-reported effects of age and gender on social functioning in a large clinical sample of children and adolescents with ADHD, autism, or co-occurring ADHD. In a previous study (Ragnarsdottir et al., 2018) we examined the interaction effects of gender and age differences on peer problems and prosocial behavior among younger children with ADHD only. The focus of the current study includes a much larger clinical group aged 5 to 18 of autistic children and adolescents and youth with co-occurring ADHD and autism, a group that has been underrepresented in clinical research. Based on previous findings (e.g., Mikami et al., 2019; Yamawaki et al., 2020), it is expected that the group with ADHD will experience the most severe peer problems compared with the other groups. The autism-only group is expected to show less prosocial behavior, according to both parents and teachers, compared with the ADHD group and co-occurring ADHD and autism group. It is also expected that the group with co-occurring ADHD and autism will show the most significant impairment and have more peer problems and less prosocial behavior in comparison to the groups diagnosed only with autism or ADHD since earlier research has indicated more impairment for this group overall (Factor et al., 2017; Rao & Landa, 2014; Sikora et al., 2012; Rosello et al., 2022).

It is also expected that girls will experience more social difficulties overall, as previous studies on girls with ADHD have suggested (e.g., Mikami et al., 2013; Mikami & Lorenzi, 2011). A significant interaction effect of age and gender is also expected to emerge for peer problems and prosocial behavior, as earlier research on youth with ADHD has shown (Ragnarsdottir et al., 2018). Whether similar interaction effects of age and gender will emerge for the groups of autistic youth and youth with co-occurring ADHD and autism will also be examined.

Method

Participants

Medical records of children and adolescents who were referred from all over Iceland for a neurodevelopmental assessment of ADHD and/or autism at the Center for Child Development and Behavior (CCDB) in Iceland (currently the Children's Mental Health Center for the Primary Health Care of the Capital Area) were examined. Referrals came from professionals within the healthcare system, school psychologists, or social services, in collaboration with parents. Still, parents were not allowed to self-refer their children to the CCDB for a neurodevelopmental assessment. The referral and evaluation process was free of charge and thus accessible to all families independent of socioeconomic status. In Iceland, the primary health care system monitors children's developmental levels and well-being systematically from birth to six years of age. Additionally, they see a school nurse for a standardized health assessment at six, nine, and twelve upon entering elementary school. Children´s developmental levels are also monitored by school services in kindergarten and elementary school. Approximately 86% of children in Iceland attend kindergarten (Hagstofa Íslands, 2024) and all children in Iceland attend elementary school since homeschooling is not allowed.

Inclusion criteria for the current study were the following: (i) meeting diagnostic criteria for ADHD, autism, or both disorders, (ii) being between the ages of 5 and 18 years, (iii) having a full-scale intelligence quotient (IQ) score of 70 or higher (or a verbal comprehension score of 70 if the full-scale score was not available) on the Wechsler Intelligence Scale for Children-IV (WISC-IV) or the Wechsler Preschool Primary Scale of Intelligence (WPPSI-R) and (iv) having the Strengths and Difficulties Questionnaire (SDQ; Goodman, 1997) completed by at least one parent or teacher. If the children had been assessed at a younger age, only the most recent assessment was included in the current study. The final sample included 2132 children and adolescents (35% girls, 65% boys; mean age = 10.07 years; *SD* = 2.98), where 223 children were diagnosed with autism (5–11-year-olds $n = 118$; 12–18-year-olds $n = 105$), and 359 children were diagnosed with co-occurring ADHD and autism (5–11-year-olds $n = 234$; 12–18-year-olds $n = 125$). For children diagnosed with ADHD ($n = 1550$) (5–11-year-olds $n = 1209$; 12–18-year-olds $n = 341$), 1078 (69%) were diagnosed with the combined presentation, 448 were diagnosed with the predominantly inattentive presentation (29%), and 24 were diagnosed with the predominantly hyperactive/impulsive presentation (2%). Children in the sample had an average IQ total score of *m (SD)* = 95.9 (14.9). Further participant characteristics are presented in Table 1.

Measures

The Strengths and Difficulties Questionnaire (SDQ; Goodman, 1997)

The SDQ (Goodman, 1997) is a 25-item screening questionnaire that measures children's strengths and weaknesses. The SDQ has five subscales: Emotional symptoms, Conduct problems, Hyperactivity, Peer problems, and Prosocial

Table 1 Participant characteristics

		Total ($n=2132$)	ADHD ($n=1550$)	Autism ($n=223$)	ADHD + autism ($n=359$)
Gender		740 (35)	572 (37)	66 (30)	102 (28)
Girls, *n* (%)		1392 (65)	978 (63)	157 (70)	257 (72)
Boys, *n* (%)		10.07 (2.98)	9.72 (2.89)**	11.57 (2.98)**	10.86 (2.94)**
Age, *m* (SD)					
Age range		5–18	5–18	5–18	5.2–18
C-GAS, *m* (SD)		60.59 (5.99)	62.61 (5.29)**	55.95 (4.42)	54.95 (4.08)
Number of other co-occurring psychiatric disorders, *n* (%)[c]					
	One	617 (28.9)	443 (28.6)	72 (32.3)	102 (28.4)
	Two	98 (4.6)	74 (4.8)	7 (3.1)	17 (4.7)
	Three	13 (0.6)	9 (0.6)	3 (1.3)	1 (0.3)

[a]Total WISC-IV or WPPSI-R score. [b]WISC-IV Verbal Comprehension Index or WPPSI-R Verbal Comprehension Index. [c]Disorders other than ADHD or autism, such as anxiety disorders, depressive disorders, oppositional defiant disorder, conduct disorder, obsessive-compulsive disorder, Tourette's or tic disorder

*$p<.05$, **$p<.01$

behavior. In the present study, only the Prosocial behavior and Peer problems subscales scores were used to assess children's social functioning and behavior. Higher scores on the Prosocial behavior subscale and lower scores on the Peer problems subscale indicate better social skills and functioning.

The scale has been translated into Icelandic, normed, and has good psychometric properties (Hrafnsdóttir, 2005; Skarphéðinsson & Magnússon, 2008). Gender and age-specific normative t-scores were used in this study. A t-score of 65 or higher is considered above the clinical cutoff for Peer problems, and a t-score of 35 or lower is the clinical cutoff score for Prosocial behavior.

The Children's Global Assessment Scale (C-GAS; Shaffer, 1983)

C-GAS is a numeric scale clinicians use to rate the general function of children aged 18 and younger. Values range from 1 to 100, where lower scores indicate more functional impairment and higher scores indicate a better function in the last month (Shaffer, 1983). The reliability of the C-GAS score is moderate (Hanssen-Bauer et al., 2007; Rey et al., 1995).

Procedure

Medical records of youth were examined for children referred to the CCDB during 2013–2021 for a neurodevelopmental assessment for ADHD and/or autism. The children had previously seen a school psychologist who administered the WISC-IV^{IS} (Salvarsdóttir & Skúlason, 2006; Wechsler, 2003) or the WPPSI-R^{IS} (Guðmundsson & Ólafsdóttir, 2003; Wechsler, 1989) depending on the children´s age. Parents of all children in the sample and their teachers (only for children ages 5–16 years) were asked to complete a few questionnaires as part of the diagnostic process, including the Icelandic version of The Strengths and Difficulties Questionnaire (SDQ; Goodman, 1997; Skarphéðinsson & Magnússon, 2008). As part of the diagnostic process, parents of all youth undergoing a neurodevelopmental assessment were administered a diagnostic interview to assess and rule out co-occurring disorders. The Kiddie Schedule for Affective Disorders and Schizophrenia (K-SADS; Kaufman et al., 1997; Kaufman et al., 2016) and/or the Anxiety Disorders Interview Schedule (ADIS-C/P; Silverman & Albano, 1996) were administered to all parents and the youth themselves if they were over ten years of age. The Icelandic version of K-SADS-PL DSM-5 has recently been shown to have good psychometric properties, such as inter-rater reliability (Hannesson et al., 2023). Children who were diagnosed with an autism spectrum disorder were administered the Autism Diagnostic Observation Schedule Test-Second Edition (ADOS-2; Lord et al., 2012) during the assessment process. Additionally, the parents were interviewed extensively about the child´s developmental history using the Autism Diagnostic Interview (ADI-R; Lord et al., 1994) and/or the Social Communication Questionnaire (Berument et al., 1999; Rutter et al., 2003) All diagnoses were finalized according to ICD-10 diagnostic criteria (World Health Organization, 2016) during a consensus meeting with a psychologist and pediatrician/child psychiatrist. The child's overall functioning was rated using the Children´s Global Assessment Scale (C-GAS; Shaffer et al., 1983).

The study is a part of the NICE project: Characteristics of neurodiverse youth in Iceland, an extensive research series of children with neurodevelopmental disorders. The study was approved by the National Bioethics Committee in Iceland (VSNb2022010022/03.01) and the Ethical Committee of the Primary Health Care of the Capital Area.

Statistical Analyses

All statistical analysis was managed using Stata 18. One-way ANOVAs were run to examine group differences in age and C-GAS scores. Tukey post-hoc tests were used to determine which group averages were statistically different. Hierarchical regression analyses were used to examine the impact of the diagnostic group (with the ADHD group as the reference category), age, gender, and the interaction of age and gender on peer problems and prosocial behavior based on parent and teacher reports.

Results

Descriptive Statistics

To examine group differences in participant characteristics, several one-way ANOVAs were performed. A statistically significant age difference between groups was found ($F(2,2129) = 53.9$, $p < .001$). A Tukey post hoc analysis revealed significant age differences between all groups. The ADHD-only group received a diagnosis at the youngest age, while youth diagnosed with autism only were, on average, the oldest when they received a diagnosis (see Table 1). The groups also differed in their overall functioning on the C-GAS scale ($F(2,2025) = 430.4$ $p < .001$). Tukey's post hoc test revealed significantly higher C-GAS scores for the ADHD-only group, indicating better functioning for the ADHD group compared with the autism groups (see Table 1). All three groups (ADHD, autism, and ADHD + autism) had comparable numbers of co-occurring psychiatric disorders, where just over one-third of all participants had at least one co-occurring disorder (see Table 1). Completion of the primary measure of the study, the SDQ, was excellent. Parent reports were available for 92% of the participants, and teacher reports were available for 91% (see Table 2 for means and standard deviations of the SDQ).

Gender and Age Effects on Peer Problems and Prosocial Behavior

To examine the impact of group membership, age, and gender on social difficulties, hierarchical regression analyses were performed on the four dependent variables (parent and teacher reports of Peer problems and Prosocial behavior); see Tables 3 and 4.

Peer Problems

For parent reports, the Autism ($p < .01$) and the ADHD + autism ($p < .01$) groups both scored higher than the ADHD group, indicating greater peer problems. Furthermore, a post hoc analysis showed that the ADHD + autism group scored significantly higher than the Autism group ($p < .05$). The results also showed that girls scored significantly higher than boys ($p < .01$), indicating they experienced greater peer problems.

For teacher reports, there was a similar pattern, with the Autism ($p < .01$) and ADHD + autism ($p < .01$) groups both scoring higher than the ADHD group and the ADHD + autism group being the highest ($p < .05$). Note that the difference between the Autism group and the ADHD + autism group is not significant once age was accounted for in the hierarchical regression (steps 3 and 4 in Table 3). Again, girls scored significantly higher than boys ($p < .01$). Here, age is also related to Peer problems such that the scores were lower for older participants ($p < .01$). The final step in the hierarchical regressions (Step 4 in Table 3) showed a significant gender-by-age interaction ($p < .01$), indicating that the impact of age on Peer problems was greater for girls compared to boys. Figure 1 illustrates the interaction effect.

Prosocial Behavior

For parent reports, the Autism ($p < .01$) and the ADHD + autism ($p < .01$) groups both scored lower than

Table 2 Means and standard deviations (t-scores) for peer problems and prosocial behavior subscales on SDQ for all groups divided by self-reported gender

Group	ADHD			Autism			ADHD + autism		
	Total	Boys	Girls	Total	Boys	Girls	Total	Boys	Girls
	($n = 1550$)	($n = 978$)	($n = 572$)	($n = 223$)	($n = 157$)	($n = 66$)	($n = 359$)	($n = 257$)	($n = 102$)
Peer problems									
Parents	57.77 (13.88)	56.39 (13.78)	60.14 (13.74)	69.51 (13.39)	68.61 (12.98)	71.88 (14.30)	73.54 (14.22)	72.75 (13.72)	75.59 (15.35)
Teachers	56.01 (12.80)	55.08 (11.70)	57.60 (14.36)	65.73 (13.07)	65.81 (11.93)	65.54 (15.81)	67.61 (12.98)	66.22 (11.88)	71.27 (14.98)
Prosocial behavior									
Parents	43.25 (14.17)	44.43 (13.84)	41.24 (14.51)	34.46 (16.13)	35.31 (15.11)	32.24 (18.52)	34.74 (15.98)	35.15 (15.06)	33.64 (18.23)
Teachers	40.64 (13.11)	41.88 (11.80)	38.53 (14.85)	33.67 (11.93)	34.85 (10.26)	30.57 (15.18)	36.09 (13.54)	38.15 (11.80)	30.72 (16.17)

Table 3 Results of hierarchical regression analyses predicting peer problems on the SDQ

		Parents[a]				Teachers[b]			
	Variable	β	R^2	*95% conf. int.*		β	R^2	*95% conf. int.*	
1	Autism	12.60**	0.17	10.54	14.66	9.36**	0.13	7.47	11.25
	ADHD + autism	15.33**		13.64	17.01	12.01**		10.45	13.57
2	Autism	12.90**	0.18	10.86	14.95	9.56**	0.13	7.67	11.44
	ADHD + autism	15.64**		13.97	17.32	12.23**		10.69	13.79
	Gender	3.57**		2.29	4.86	2.46**		1.26	3.66
3	Autism	12.91**	0.18	10.82	14.99	10.62**	0.15	8.71	12.54
	ADHD + autism	15.64**		13.95	17.33	12.83**		11.26	14.40
	Gender	3.58**		2.28	4.87	2.75**		1.55	3.95
	Age	-0.00		-0.21	0.21	-0.57**		-0.78	-0.36
4	Autism	12.86**	0.18	10.77	14.95	10.85**	0.15	8.94	12.77
	ADHD + autism	15.64**		13.94	17.33	12.80**		11.24	14.37
	Gender	2.11		-2.44	6.66	10.29**		5.78	14.79
	Age	-0.05		-0.32	0.21	-0.31*		-0.57	-0.05
	Gender x age	0.14		-0.28	0.57	-0.76**		-1.19	-0.32

Note: The ADHD group is the reference category

[a]Step: (1) $F_{2,1968} = 203.80^{**}$. (2) $F_{3,1967} = 147.78^{**}$. (3) $F_{4,1966} = 210.78^{**}$. (4) $F_{5,1965} = 203.80^{**}$

[b]Step: (1) $F_{2,1945} = 141.77^{**}$. (2) $F_{3,1944} = 100.60^{**}$. (3) $F_{4,1943} = 83.30^{**}$. (4) $F_{5,1942} = 69.32^{**}$

*$p < .05$; **$p < .01$

Table 4 Results of hierarchical regression analyses predicting prosocial behavior on the SDQ

		Parents[a]				Teachers[b]			
	Variable	β	R^2	*95% conf. int.*		β	R^2	*95% conf. int.*	
1	Autism	-9.16**	0.07	-11.31	-7.01	-6.79**	0.03	-8.72	-4.85
	ADHD + autism	-8.45**		-10.21	-6.69	-4.48**		-6.07	-2.89
2	Autism	-9.44**	0.08	-11.58	-7.30	-7.09**	0.05	-8.72	-4.85
	ADHD + autism	-8.75**		*-10.50*	*-6.99*	-4.82**		-6.07	-2.89
	Gender	-3.36**		-4.71	-2.02	-3.85**		-5.08	-2.63
3	Autism	-9.84**	0.08	-12.02	-7.66	-8.16**	0.07	-10.11	-6.21
	ADHD + autism	-8.99**		*-10.77*	*-7.22*	-5.41**		-7.00	-3.82
	Gender	-3.50**		-4.86	-2.15	-4.14**		-5.36	-2.92
	Age	0.21		-0.00	0.04	0.57**		0.35	0.79
4	Autism	-9.75**	0.08	-11.93	-7.57	-8.36**	0.07	-10.31	-6.41
	ADHD + autism	-8.99**		*-10.76*	*-7.22*	-5.38**		-6.97	-3.80
	Gender	-0.53		-5.30	4.25	-10.53**		-15.12	-5.94
	Age	0.32*		0.04	0.59	0.35**		0.09	0.62
	Gender x age	-0.29		-0.74	0.16	0.64**		0.20	1.09

Note: The ADHD group is the reference category

[a]Step: (1) $F_{2,1962} = 68.95^{**}$. (2) $F_{3,1961} = 54.48^{**}$. (3) $F_{4,1960} = 41.78^{**}$. (4) $F_{5,1959} = 33.76^{**}$

[b]Step: (1) $F_{2,1938} = 33.94^{**}$. (2) $F_{3,1937} = 35.78^{**}$. (3) $F_{4,1936} = 33.85^{**}$. (4) $F_{5,1935} = 28.79^{**}$

*$p < .05$; **$p < .01$

the ADHD group, indicating less Prosocial behavior. There was not a significant difference between the Autism and the ADHD + autism group. Girls scored significantly lower than boys ($p < .01$), indicating they demonstrated less Prosocial behavior (see Table 4).

For teacher reports, there was a similar pattern again. The Autism ($p < .01$) and ADHD + autism ($p < .01$) groups both scored lower than the ADHD group. The ADHD + autism group had the lowest score ($p < .05$), though the difference between the Autism group and the ADHD + autism group was marginally significant in steps 1 and 2 (see Table 4). Again, girls scored significantly lower ($p < .01$). Age was also related to Prosocial behavior, with older participants receiving higher scores ($p < .01$). As illustrated in Fig. 2, a gender-by-age interaction was found ($p < .01$). The interaction indicates the impact of age on Prosocial behaviors was greater for girls compared to boys.

To examine whether this pattern of results was exclusively found among children and adolescents with ADHD and/or autism, the same hierarchical regression analyses

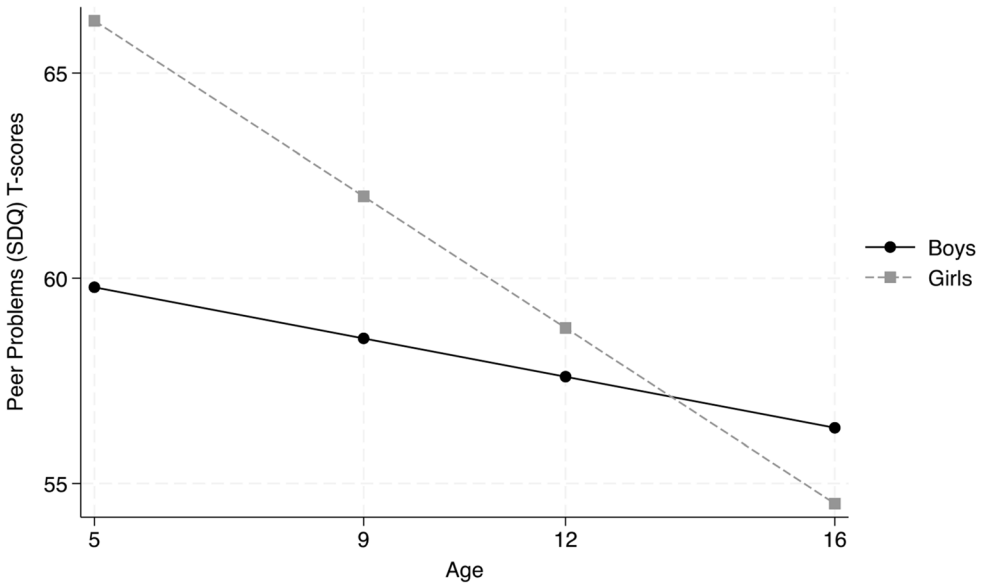


Fig. 1 Interaction of gender and age based on teacher reports of Peer problems

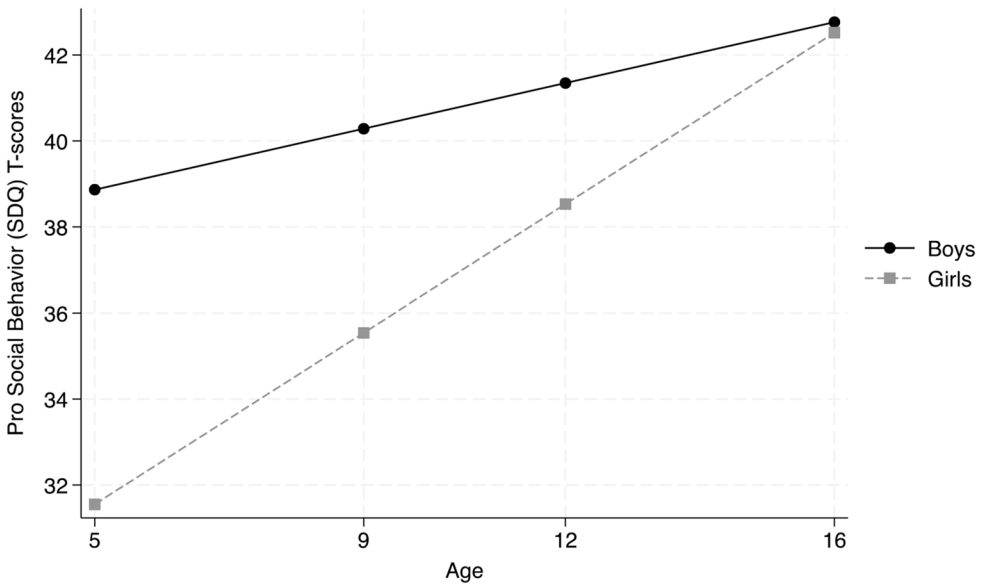


Fig. 2 Interaction of gender and age based on teacher reports of Prosocial behavior

were performed for comparison on the four dependent variables (parent and teacher reports of Peer problems and Prosocial behavior) for a separate dataset of youth referred to the CCDB for an ADHD or an autism assessment process but who did not meet diagnostic criteria for ADHD or autism ($n = 320$; 38% girls, 62% boys; mean age = 9.53 years; SD = 2.44). The four models — each using age, gender, and an age-by-gender interaction as predictors — were all found to be non-significant, indicating that this pattern of results only emerged for children and adolescents diagnosed with ADHD, autism or both.

Discussion

This study aimed to examine peer problems and prosocial behavior from parents' and teachers' perspectives in a large clinical nationwide sample of youth with ADHD and/or autism. It was expected that children with co-occurring ADHD and autism would have the greatest impairment in social functioning when compared to youth with only ADHD or only autism, as some previous studies have indicated (e.g., Factor et al., 2017; Rosello et al., 2022). The results only partly supported this hypothesis. Parents rated the group with co-occurring ADHD and autism to have significantly more peer problems compared with youth with ADHD only or autism only. However, teachers reported similar levels of peer problems for autistic youth and youth

with co-occurring ADHD and autism, while the ADHD-only group had significantly fewer peer problems. Nevertheless, results for the prosocial behavior subscale were in line with other previous studies (e.g., Harkins et al., 2022; Yamawaki et al., 2020). Both parents and teachers reported significantly more prosocial behavior for the ADHD group compared to the autistic group and the group with co-occurring ADHD and autism. These results suggest that children with ADHD have fewer peer problems and demonstrate to a greater extent prosocial behavior when compared with autistic youth or youth with co-occurring ADHD and autism. The results also indicate that overall, social difficulties of youth with autism are rated greater by parents and teachers, independent of whether they have co-occurring ADHD or not.

Interestingly, the interaction of age and gender was only significant according to teacher reports of peer problems and prosocial behavior and not by parent reports. These results are unsurprising since informant discrepancies are usually the rule rather than the exception in clinical child research (De Los Reyes, 2011). Often, informant discrepancies arise between teachers and parents because they see children in different circumstances. Different situations call for different behavior demands, and raters from various settings have been found to have a lower agreement than raters from the same setting (Achenbach, 2006; De Los Reyes et al., 2023). This may be especially true for ratings of social behavior and peer interactions since parents usually see the child at home playing with one child or in a small group. In contrast, the teachers see children interacting in a large group setting. Previous studies in different countries on the SDQ have shown that parent and teacher agreement on the Peer problems and Prosocial behavior subscales tends to be low to moderate (Cheng et al., 2018). It has been argued that parent and teacher discrepancies reflect more than a measurement error but can provide valuable insights into how children behave differently depending on who they are interacting with within different settings. Such information can be used to improve treatment effects (De Los Reyes, 2011).

In terms of gender differences, the results showed that girls had greater peer problems than boys, according to both teachers and parents, and girls showed less prosocial behavior. The hypothesis that girls would have more social difficulties than boys was supported and in line with previous studies (e.g., Mikami et al., 2013; Ragnarsdottir et al., 2018). However, significant interaction effects of age and gender were only found for teacher reports of peer problems and prosocial behavior. While it could be assumed beforehand that youth being referred for a neurodevelopmental assessment at a younger age would likely be more impaired than those being referred during adolescence, this pattern of results only emerged for girls and not for boys. According to teachers, younger girls experienced more social difficulties than older girls, while older and younger boys experienced similar levels of problems with peers. The same interaction effect was seen for prosocial behavior, where teachers reported that younger girls showed less prosocial behavior than older girls. These results suggest that girls with neurodevelopmental disorders struggle to a greater extent with social functioning compared to boys, especially younger girls.

It has been suggested that girls with neurodevelopmental difficulties, like ADHD and/or autism, may seem to stand out as more deviant in a group of girls. At the same time, there may be more variability in groups of boys, so boys with neurodevelopmental disorders may blend in more easily (Mikami et al., 2013; Mikami & Lorenzi, 2011). It is unclear why younger girls had greater difficulties than older girls in this particular study. Research has shown that higher-functioning girls with autism may be identified later than boys and that the average time between initial concern and a diagnosis of autism may be longer for girls than boys (Begeer et al., 2013). Perhaps older girls have gotten better at using compensatory behaviors to mask their social challenges, as other research has suggested (Attwood, 2008; Cook et al., 2021; Dean et al., 2017). It is also possible that the older girls have found common ground with others regarding their unique interests and are now part of a peer group that shares their interests, thus experiencing fewer peer problems and showing increased prosocial behavior. Since this is a cross-sectional study, this issue remains unanswered, and longitudinal data would be necessary to answer these questions.

Strengths and Limitations

The current study has many strengths. First, the results are based on a large, nationwide sample of boys and girls who completed an extensive and interdisciplinary in-person assessment protocol for their ADHD and/or autism diagnoses. Results for these youth with neurodevelopmental disorders are based on synchronized standard assessment protocols and not just on questionnaire reports for symptoms of ADHD and/or autism, as in some previous studies (e.g., Factor et al., 2017). Second, the sample consisted of youth referred for various neurodevelopmental difficulties, that is, either ADHD symptoms, autism, or both. In similar studies, the clinical samples have been referred mainly for an autism evaluation (e.g., Harkins et al., 2022), thus missing out on essential comparisons with a more basic clinical group of youth with only ADHD. Fourth, this is one of the few clinical samples where 35% of the youth with neurodevelopmental disorders are females. Few such studies are available on girls with co-occurring ADHD and autism, and

some studies have focused only on boys (e.g., Yamawaki et al., 2020). Finally, while many previous studies have focused mainly on parent-reported social functioning (Mikami et al., 2019), the current study provides valuable insight into how teachers view these children's social functioning. Missing information from teachers was minimal since over 90% of teachers for the current sample completed the SDQ for these youth with neurodevelopmental disorders. Knowing that young girls with co-occurring ADHD and autism are struggling socially in the school environment adds meaningfully to our understanding of gender differences in children with these disorders. Identifying how behavior appears in different situations provides valuable insights into where more support is needed and at what developmental stage appropriate interventions should be a priority.

The current study also has some limitations. First, only one questionnaire was administered to assess social functioning and behavior. Additional information from direct observations would have been valuable for this study. Second, social functioning evaluated in terms of peer problems and prosocial behaviors reported only by parents and teachers tells us very little about how the children view their social difficulties or how their peers evaluate them. It also tells us very little about the number and quality of friendships youth with neurodevelopmental disorders may have, which are essential areas of research (e.g., Petrina et al., 2014; Spender et al., 2023), especially for developing effective interventions. Third, although the predictive models are significant, they explain a relatively low proportion of the variance. Adding additional variables into our data collection and models would have been beneficial since other factors such as low SES (Becker et al., 2015), maternal education, parents psychological well-being and being in a reconstituted family (McMunn et al., 2001), having fewer adaptive skills and more severe autism symptoms (Hastings et al., 2022) have all been associated with worse outcomes on the SDQ questionnaire. Finally, the clinical group consisted only of youth with no intellectual impairment. While this makes it easier to generalize the results to similar groups of youth, further research is needed to examine whether the same patterns of social functioning emerge for youth with neurodevelopmental disorders and intellectual impairment. Longitudinal studies would be valuable in exploring how social difficulties develop differently between boys and girls throughout various developmental stages.

Conclusion

This study adds to the existing literature on gender differences in social functioning and social behaviors of youth with neurodevelopmental disorders. The study also adds to the sparse literature on characteristics of youth with co-occurring ADHD and autism. The study is clinically relevant since it adds insights into how treatment for girls and boys might be developed differently to address their social difficulties at different developmental stages. The results also highlight that autistic youth struggle more with social difficulties compared with youth with ADHD only and that younger girls have more social difficulties compared with older girls.

In the future, longitudinal research of peer problems and prosocial behavior would help establish the developmental trajectories of those social difficulties of youth with ADHD and/or autism, as well as shed light on gender and age differences of those difficulties across different developmental stages. The current study does, however, indicate that it might be important to support youth with neurodevelopmental disorders, especially younger girls, in the school setting. These findings could help clinicians develop appropriate social interventions that might benefit groups of children with neurodevelopmental disorders at a certain age, for example, during the first years of elementary school. Such intervention programs, if successful in minimizing peer problems and increasing prosocial behaviors, could, in turn, impact youth with neurodevelopmental disorders developing further problems. Since various co-occurring disorders, such as anxiety and depression, are more common among youth with neurodevelopmental disorders (e.g., Gordon-Lipkin et al., 2018; Tung et al., 2016), it is important to consider whether appropriate social interventions at the appropriate developmental stage could reduce the chances of developing such co-occurring disorders.

Acknowledgements Our sincerest gratitude goes to the families and teachers who completed measures for our research.

Author contribution FJG, BR, UN and DKH conceived the research question and design and wrote the first draft of the article. FH ran all the analyses but FJG and DKH had full access to all the data in the study. All authors reviewed drafts, edited the article, and approved of the final draft.

Funding No funding was received for conducting this study. The authors have no relevant financial or non-financial interests to disclose.

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